

Problem 10.10

Suppose \$2,000 is invested for 3 years with a constant force of interest equal to 9%.

- (a) Find the accumulation function.
- (b) Find the effective rate of interest.
- (c) Find the value of the investment after 3 years.

Solution

The constant force of interest is

$$\delta = 0.09.$$

Part (a).

For a constant force of interest, the accumulation function is

$$a(t) = e^{\delta t}.$$

Therefore,

$$\boxed{a(t) = e^{0.09t}}.$$

Part (b).

The annual effective interest rate satisfies

$$1 + i = e^{\delta}.$$

Thus,

$$i = e^{0.09} - 1.$$

Therefore,

$$\boxed{i = 0.09417428}.$$

Part (c).

The accumulated value after 3 years is

$$AV = 2000a(3).$$

Substitute the accumulation function:

$$AV = 2000e^{0.09(3)} = 2000e^{0.27}.$$

Thus,

$$\boxed{AV = 2619.93}.$$